

ASK MY DOCUMENTS

Design and Evaluation of a Transparent Retrieval-Augmented Generation System for Uploaded Documents

FINAL REPORT

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ABSTRACT

Ask My Documents is a local academic Retrieval-Augmented Generation (RAG) prototype for question answering over user-uploaded PDF, DOCX, and TXT files. The project addresses a trust problem in document-based AI: a fluent answer is not sufficient when the user needs to know whether the response is supported by private, recent, or document-specific evidence. The prototype therefore exposes document chunks, retrieval ranks, similarity scores, relevance-threshold decisions, source metadata, and answer source labels so that the path from document to answer can be inspected. The evaluation compared fixed-size and paragraph-aware chunking at retrieval depths of top-3 and top-5 using nine documents and 30 questions, including 24 answerable and six deliberately unanswerable cases. Across four configurations, the experiment produced 120 primary result rows and 480 retrieved-chunk audit rows. All configurations achieved 100% retrieval hit on answerable questions, 100% citation-label inclusion, and correct abstention in the test set. Paragraph-aware retrieval ranked the expected source first in 22 of 24 questions compared with 19 of 24 for fixed-size retrieval. Top-5 did not improve retrieval recall, but increased average answerable context from 2.81 to 4.48 chunks and increased non-gold-source context from 48.9% to 64.2%. A manual audit also showed that a strict keyword grader undercounted semantically acceptable answers. The strongest recommendation is therefore to use top-3 as the current retrieval baseline and improve semantic and human evaluation before selecting a final chunking strategy.

Keywords: *Retrieval-Augmented Generation, document question answering, vector retrieval, chunking, evaluation, citations, abstention, transparent AI*

TABLE OF CONTENTS

1. Introduction	1
1.1 Background and Motivation	1
1.2 Problem Statement	1
1.3 Aim, Objectives, and Scope	1
1.4 Research Questions	2
2. Literature Review	3
2.1 RAG and Dense Retrieval	3
2.2 Context Quantity and Evaluation	3
3. Project Description and Solution	4
3.1 Architecture and Workflow	4
3.2 Implemented Components and Contribution	4
3.3 Interface and Transparency Features	5
3.4 Progress and Deliverables	7
4. Evaluation Methodology	8
4.1 Corpus and Questions	8
4.2 Configurations and Metrics	8
5. Results and Analysis	10
5.1 Quality and Retrieval Results	10
5.2 Context Efficiency and Response Time	10
5.3 Analysis by Research Question	12
6. Discussion, Benefits, and Challenges	13
6.1 What the Prototype Demonstrates	13
6.2 Error Analysis and Metric Validity	13
6.3 Benefits and Learning Outcomes	13
6.4 Problems, Challenges, and Mitigations	14
6.5 Limitations	14
7. Recommendations and Future Work	15
8. Conclusion	16
References	17
Appendices	18

LIST OF TABLES

Table 1. Project scope	1
Table 2. Research questions	2
Table 3. Implemented components	4
Table 4. Build-versus-reuse decision	5
Table 5. Progress against proposed deliverables	7
Table 6. Evaluation dataset summary	8
Table 7. Experimental configurations	8
Table 8. Metric definitions	9
Table 9. Quality metrics by configuration	10
Table 10. Context efficiency and timing	10
Table 11. Examples of wording-sensitive grading failures	13
Table 12. Project challenges and mitigations	14
Table 13. Output files	18
Table 14. Corpus inventory	18

LIST OF FIGURES

Figure 1. Implemented indexing and question-answering pipelines	4
Figure 2. Document-processing interface	6
Figure 3. Retrieval-and-answer interface	7
Figure 4. Quality metrics by configuration	10
Figure 5. Non-gold-source context by configuration	11
Figure 6. Response time by configuration	11

1. Introduction

1.1 Background and Motivation

Large language models can produce useful and fluent responses, but their internal knowledge may not contain information that exists only in a user's private files or the latest version of a policy, procedure, or technical document. Retrieval-Augmented Generation (RAG) addresses this limitation by retrieving external passages and conditioning generation on that evidence. Lewis et al. introduced RAG as a combination of parametric model knowledge and non-parametric retrieved memory, with provenance and knowledge updating as important motivations [1].

Ask My Documents applies this idea to uploaded documents. A user provides PDF, DOCX, or TXT files, the application extracts and indexes their text, and a question triggers similarity search over document chunks. Only passages that satisfy the relevance policy are passed to the language model, and the answer is displayed together with source labels and inspectable evidence. The project is deliberately framed as an academic prototype rather than a production document-management platform.

1.2 Problem Statement

Managed file-search systems can simplify development, but they may hide chunk boundaries, ranking scores, threshold behaviour, source selection, and other retrieval decisions. When these decisions are hidden, a developer can observe whether an answer looks reasonable but may have difficulty explaining why the answer was produced or whether the correct evidence was used. The project therefore treats transparency as a design requirement: retrieval decisions must be visible enough to investigate both success and failure.

Core project motivation

A fluent answer is not automatically a verified answer. For document question answering, users should be able to inspect the evidence path from source document to retrieved passage to cited response.

1.3 Aim, Objectives, and Scope

The project aim is to design, implement, and evaluate a small configurable RAG system that answers questions using evidence retrieved from user-uploaded documents. The work focuses on design choices that directly affect retrieval quality, grounding, and inspectability rather than on training new models.

Scope category	Description
Included	Local Streamlit prototype; PDF, DOCX, and TXT extraction; fixed-size and paragraph-aware chunking; Pinecone indexing and vector search; top-k and relevance-threshold controls; grounded generation; visible source labels; automated evaluation and audit outputs.
Not included	User authentication; production deployment; model training; advanced OCR for scanned documents; complex table or diagram understanding; enterprise security controls; large-scale statistical validation.

Table 1. Project scope.

The original proposal described local storage and direct similarity calculation as the initial retrieval design. During implementation, the vector-search component evolved to Pinecone integrated vector search. This was a

build-versus-reuse change rather than a change in research purpose: chunking strategy, source metadata, top-k ranking, thresholds, citations, and evaluation remained under project control and visible in the audit trail.

1.4 Research Questions

ID	Research question
RQ1	How does document chunking strategy affect retrieval hit rate, expected-source ranking, automated answer coverage, and citation inclusion?
RQ2	How does retrieval depth, specifically top-3 versus top-5 chunks, affect answer quality, response time, and non-gold-source context?
RQ3	How reliably can a relevance threshold and grounded prompt cause the system to abstain when the uploaded documents do not contain an answer?

Table 2. Research questions guiding implementation and evaluation.

These questions separate retrieval behaviour from answer-generation behaviour. That distinction is important because a system may retrieve the correct source and still generate a weak answer, or it may display a citation label without establishing that every claim is supported by the cited text.

2. Literature Review

2.1 Retrieval-Augmented Generation and Dense Retrieval

RAG systems combine a generation model with an external retrieval mechanism. In the original RAG formulation, a sequence-to-sequence model is paired with a dense vector index so that generated output can be conditioned on retrieved passages [1]. This separation is valuable for document question answering because external evidence can be changed or replaced without retraining the language model, and the retrieved passages can provide provenance for the final response.

Dense passage retrieval provides a practical basis for semantic search over text chunks. Karpukhin et al. showed that dense representations can be used effectively for passage retrieval in open-domain question answering [2]. Ask My Documents uses the same general idea at a smaller scale: document chunks are represented in a vector-search system, and the user question is used to retrieve semantically similar chunks. The project does not claim novelty in embedding or retrieval models; the contribution is in making the surrounding workflow inspectable and experimentally configurable.

2.2 Context Quantity, Grounding, and RAG Evaluation

More retrieved context is not automatically better. Liu et al. demonstrated that language models do not use long contexts uniformly and can be sensitive to the position of relevant information [3]. This motivates an explicit comparison of top-3 and top-5 retrieval rather than assuming that additional passages will improve answer quality. In a document system, additional context can be useful when evidence is distributed across sections, but it can also introduce distraction, ambiguity, or unnecessary token use.

RAG evaluation is also multidimensional. RAGAs distinguishes the quality of retrieval, faithfulness to evidence, and generation quality [4]. ARES likewise evaluates context relevance, answer faithfulness, and answer relevance [5]. These frameworks support the evaluation philosophy used here: retrieval hit, expected-source rank, answer coverage, citation-label inclusion, abstention, context efficiency, and latency are measured separately rather than collapsed into one score.

Literature-review implication

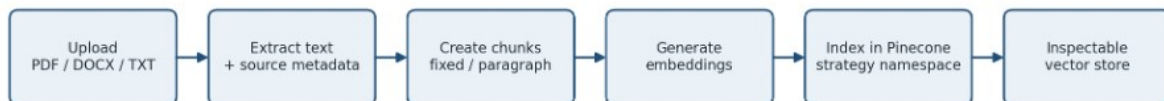
A single exact-match or keyword score cannot fully characterize a RAG system. Retrieval can succeed while a wording-sensitive answer metric fails; similarly, a citation label can be present without proving claim-by-claim entailment.

3. Project Description and Solution

3.1 Architecture and End-to-End Workflow

The implemented solution separates document indexing from question answering. During indexing, the system extracts text and source metadata, creates document chunks, and stores searchable vector representations in strategy-specific Pinecone namespaces. During question answering, the system retrieves ranked chunks, applies a relevance threshold, constructs a grounded prompt, sends the selected evidence to the language model, and returns an answer with source labels. An audit trail records the passages, metadata, scores, ranks, threshold decisions, and citations used in evaluation.

A. Document indexing pipeline



B. Question-answering pipeline

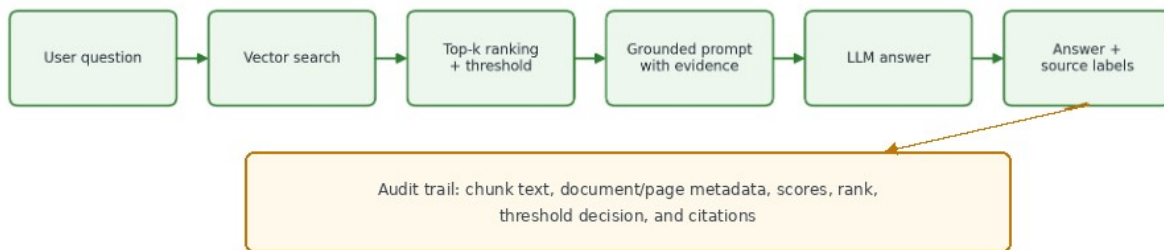


Figure 1. Implemented indexing and question-answering pipelines, including the audit trail used for evaluation.

3.2 Implemented Components and Project Contribution

Component	Project location	Purpose
Document extraction	rag/extractors.py	Extracts page- or paragraph-level text from TXT, PDF, and DOCX while preserving source metadata.
Chunking	rag/chunking.py	Creates fixed-size overlapping chunks and paragraph-aware chunks.
Indexing and search	rag/embeddings.py	Uses Pinecone integrated embedding/vector search with separate namespaces for each chunking strategy.
Retrieval filtering	rag/retriever.py	Applies top-k retrieval, relevance thresholding, source numbering, and citation preparation.
Grounded generation	rag/generator.py; rag/prompts.py	Instructs the model to answer from retrieved passages and cite displayed source labels.
Evaluation	rag/evaluator.py	Runs the four experimental configurations and writes result and chunk-level audit files.

Table 3. Implemented components and responsibilities.

The project does not claim to create a new embedding model or language model. Its contribution is the transparent workflow and controlled evaluation around existing services. The work therefore emphasizes the software and experimental decisions that can be explained and reproduced: segmentation, metadata, namespace separation, retrieval settings, prompt construction, evidence display, abstention behaviour, and experiment logging.

Category	Details
Implemented for the project	Chunking logic; metadata and citation mapping; retrieval configuration; threshold filtering; prompt construction; Streamlit workflow; evaluation questions and expected sources; metrics; CSV audit outputs.
Reused appropriately	Document-reading libraries; Streamlit; Pinecone vector search; external embedding and language-model services; Pandas and Matplotlib for analysis.
Reproducibility gap in exported artifacts	Exact embedding and generation model identifiers and some provider settings were not preserved in the supplied evaluation exports. These should be captured automatically in future runs.

Table 4. Build-versus-reuse decision.

3.3 Interface and Transparency Features

The user interface exposes the settings that influence RAG behaviour, including chunking strategy, chunk size and overlap, retrieval depth, relevance threshold, and answer length. The processed-document view confirms the selected strategy, chunk count, namespace, and indexing status, and it allows chunk preview before questions are asked.

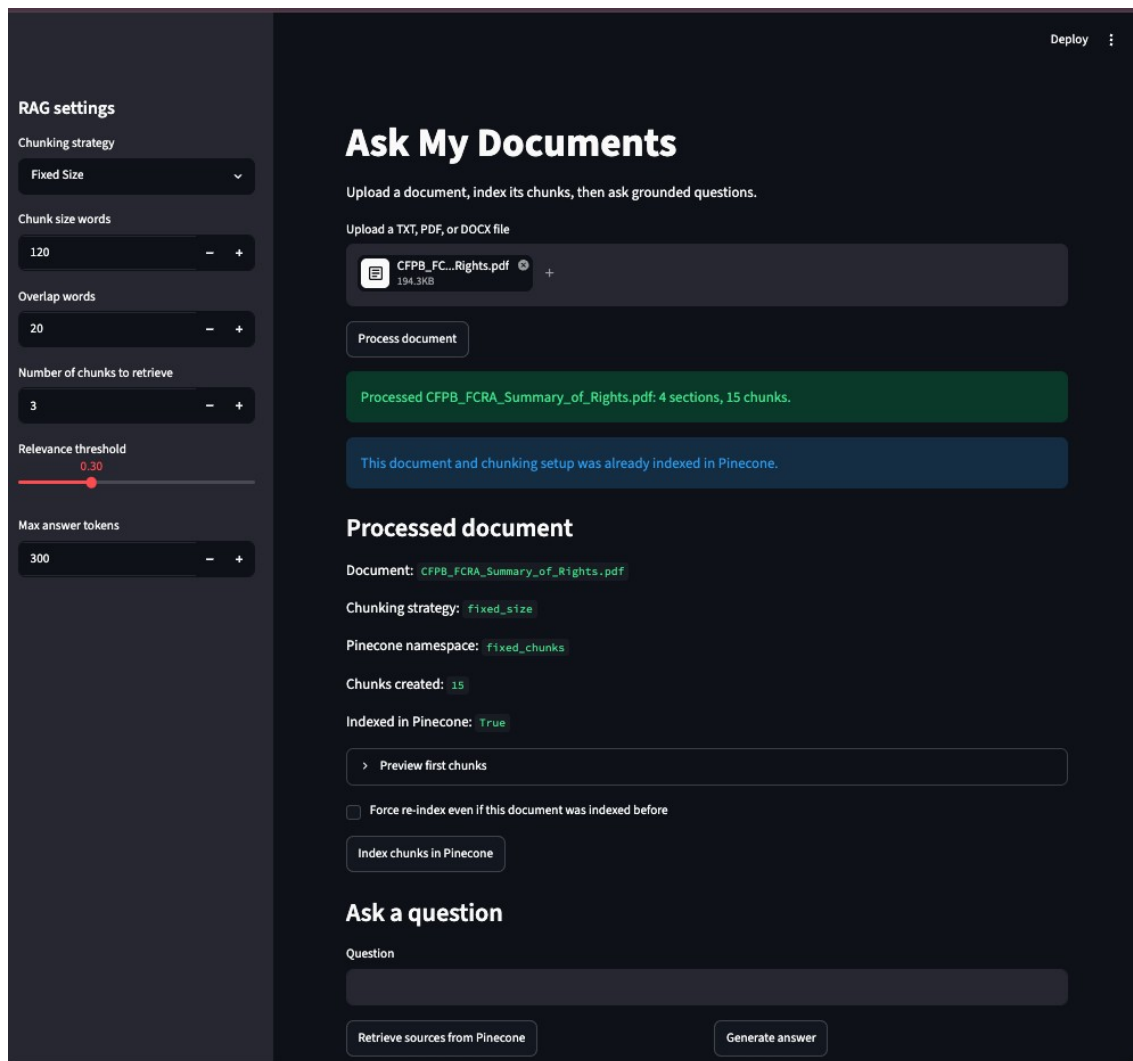


Figure 2. Document-processing interface showing configurable RAG settings, processed-document details, and chunk preview.

The retrieval-and-answer view makes the main RAG decisions inspectable. It displays raw ranked sources and scores, the generated answer, response time, and the passages actually sent to the language model after threshold filtering. This makes it possible to distinguish retrieval failure from generation failure: a developer can verify which passages were returned, which were retained, and whether the answer is consistent with the displayed evidence.

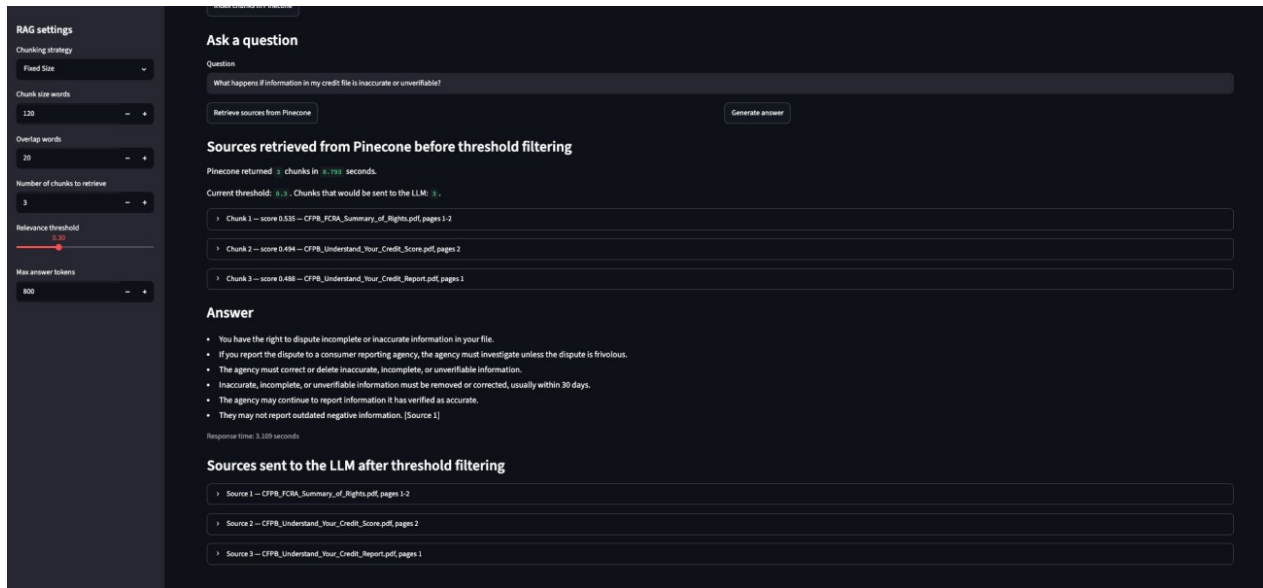


Figure 3. Retrieval-and-answer interface showing ranked sources, similarity scores, answer output, and passages sent to the language model.

3.4 Progress, Deliverables, and Accomplishments

The proposal defined a practical six-week objective: build a working document question-answering prototype, preserve source evidence, compare selected retrieval design choices, and produce evaluation results. The final project met the core deliverables while keeping production features outside scope.

Proposed objective / deliverable	Final status	Evidence of completion
Working Streamlit application	Completed	Local interface for document upload, processing, question answering, evidence display, and configuration.
PDF, DOCX, and TXT document handling	Completed	Extraction and metadata preservation implemented for all three file types.
Transparent RAG workflow	Completed	Visible chunks, scores, source labels, threshold behaviour, and audit outputs.
Compare fixed-size and paragraph-aware chunking	Completed	Both strategies indexed in separate namespaces and tested under the same question set.
Compare top-3 and top-5 retrieval	Completed	Four configurations evaluated; top-3 identified as the practical baseline.
Test unsupported-question handling	Completed for prototype test set	Six deliberately unanswerable questions used to test threshold plus grounded-prompt abstention.
Evaluation dataset and result files	Completed	30 questions, 120 primary rows, and 480 retrieved-chunk audit rows.
Production deployment and user management	Intentionally out of scope	Deferred to future work in accordance with the proposal scope.

Table 5. Progress against proposed project objectives and deliverables.

4. Evaluation Methodology

4.1 Corpus and Question Set

The evaluation corpus contains nine documents: the project proposal and public documents from the Consumer Financial Protection Bureau, Occupational Safety and Health Administration, National Institute of Standards and Technology, Federal Emergency Management Agency, and Medicare. The documents are predominantly text-heavy and factual. Several contain overlapping procedural or policy language, which creates realistic ambiguity for retrieval.

Evaluation item	Count
Documents	9
Prepared questions	30
Answerable questions	24
Unanswerable questions	6
Configurations	4
Primary result rows	120
Retrieved-chunk audit rows	480

Table 6. Evaluation dataset summary.

Each answerable question includes an expected answer field and an expected source document. The six unanswerable questions are intentionally outside the available evidence so that the system can be evaluated on whether it avoids unsupported generation rather than simply whether it produces text.

4.2 Experimental Configurations

Configuration	Chunking strategy	Pinecone namespace	Top-k
fixed_top_3	fixed_size	fixed_chunks	3
fixed_top_5	fixed_size	fixed_chunks	5
paragraph_top_3	paragraph_aware	paragraph_chunks	3
paragraph_top_5	paragraph_aware	paragraph_chunks	5

Table 7. Four experimental configurations.

All configurations used the same question set and a relevance threshold of 0.35. Holding the threshold constant allowed the experiment to isolate the effects of chunking strategy and retrieval depth. The evaluator recorded retrieved ranks and scores, threshold outcomes, final answers, source labels, abstention decisions, and response times.

4.3 Evaluation Metrics

Metric	Operational definition
Retrieval hit	At least one chunk sent to the model matched the designated expected source document.

Metric	Operational definition
Expected-source rank	Rank of the expected source among raw retrieved results. Rank 1 indicates that the expected source was returned first.
Automated keyword coverage	All semicolon-separated expected phrases appeared after punctuation normalization. This is a strict coverage check, not a semantic correctness judgment.
Citation-label inclusion	The final answer contained at least one displayed source label such as [Source 1]. It does not establish claim-by-claim entailment.
Abstention correctness	The system abstained on unsupported questions and did not abstain on answerable questions.
Non-gold-source context	A chunk sent for an answerable question that did not come from the designated expected source. Related supporting passages may still be useful.
Response time	Total retrieval plus generation time recorded per question/configuration case.

Table 8. Operational definitions of evaluation metrics.

4.4 Experimental Procedure and Validity Controls

Each of the 30 questions was executed under each of the four configurations. Primary results were written at question/configuration level, while a second audit file retained retrieved-chunk details such as rank, score, threshold result, source metadata, and whether the chunk came from the expected source. This two-level structure made it possible to explain aggregate results using the underlying evidence.

The evaluation is intentionally modest in scale. It is designed to compare system configurations under a controlled prototype setting, not to claim population-level performance. The fixed threshold, single timing observation per case, limited unanswerable set, and single embedding/generation setup are therefore treated as validity limitations rather than hidden assumptions.

5. Results and Analysis

5.1 Quality and Retrieval Results

Configuration	Retrieval hit	Expected source rank 1	Keyword coverage	Citation-label inclusion	Abstention correctness
fixed_top_3	100.0%	79.2%	83.3%	100.0%	100.0%
fixed_top_5	100.0%	79.2%	79.2%	100.0%	100.0%
paragraph_top_3	100.0%	91.7%	79.2%	100.0%	100.0%
paragraph_top_5	100.0%	91.7%	70.8%	100.0%	100.0%

Table 9. Quality metrics by configuration.

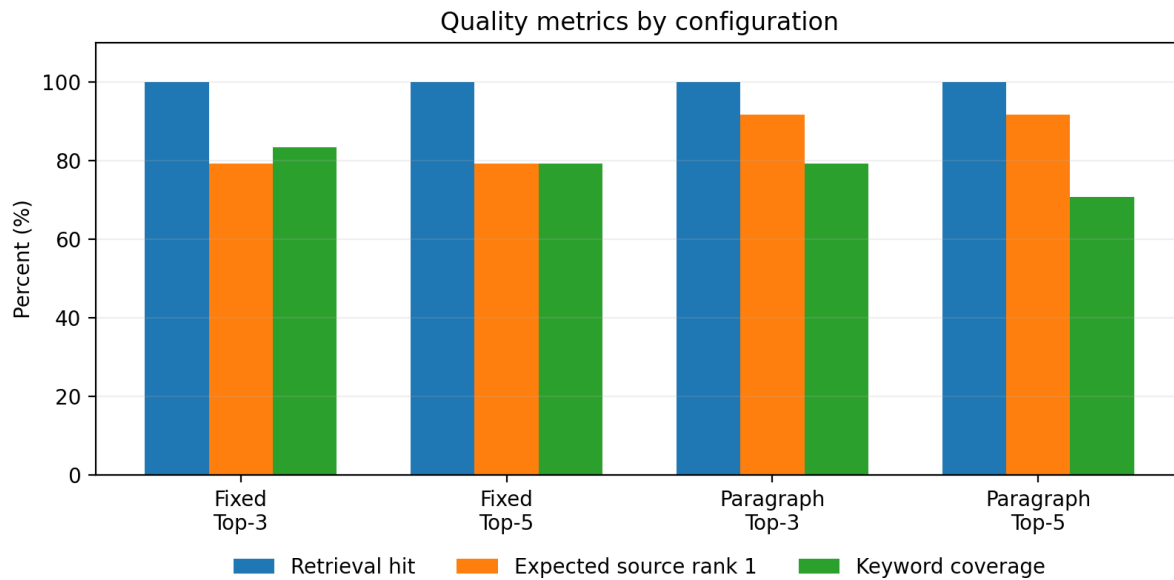


Figure 4. Retrieval hit, expected-source first-rank placement, and strict automated keyword coverage for answerable questions.

All four configurations retrieved an expected source for every answerable question. Paragraph-aware chunking placed the expected source first more often than fixed-size chunking: 22 of 24 questions, or 91.7%, compared with 19 of 24, or 79.2%. The strict keyword metric favoured fixed_top_3, but later error analysis shows that this difference is strongly affected by wording-sensitive grading.

5.2 Context Efficiency and Response Time

Configuration	Mean time (s)	Median time (s)	Avg. chunks sent (all questions)	Non-gold context (answerable only)
fixed_top_3	2.874	2.481	2.47	44.8%
fixed_top_5	2.386	2.255	3.93	57.9%
paragraph_top_3	2.381	2.361	2.53	52.9%
paragraph_top_5	2.428	2.361	4.00	70.4%

Table 10. Context efficiency and response-time measurements.

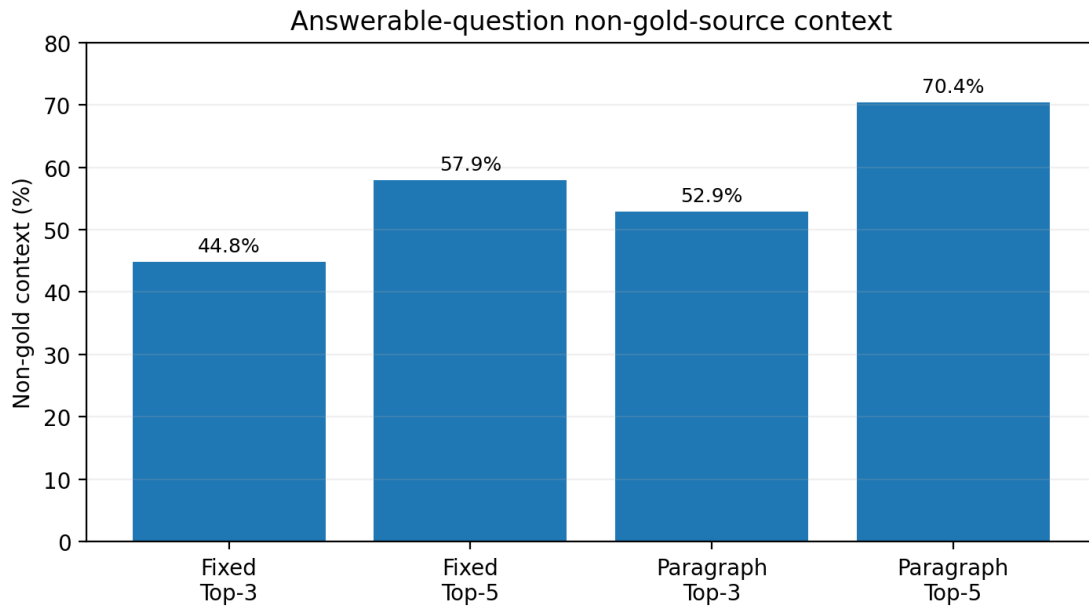


Figure 5. Answerable-question non-gold-source context. Top-5 retrieves more passages but does not improve source recall.

Top-5 increased the context sent to the model without retrieving an expected source for any additional answerable question, because top-3 had already achieved 100% retrieval hit. Across chunking strategies, top-3 sent 2.81 chunks on average for answerable questions, while top-5 sent 4.48. The non-gold-source rate increased from 48.9% to 64.2%. This is the strongest quantitative basis for preferring top-3 in the current corpus.

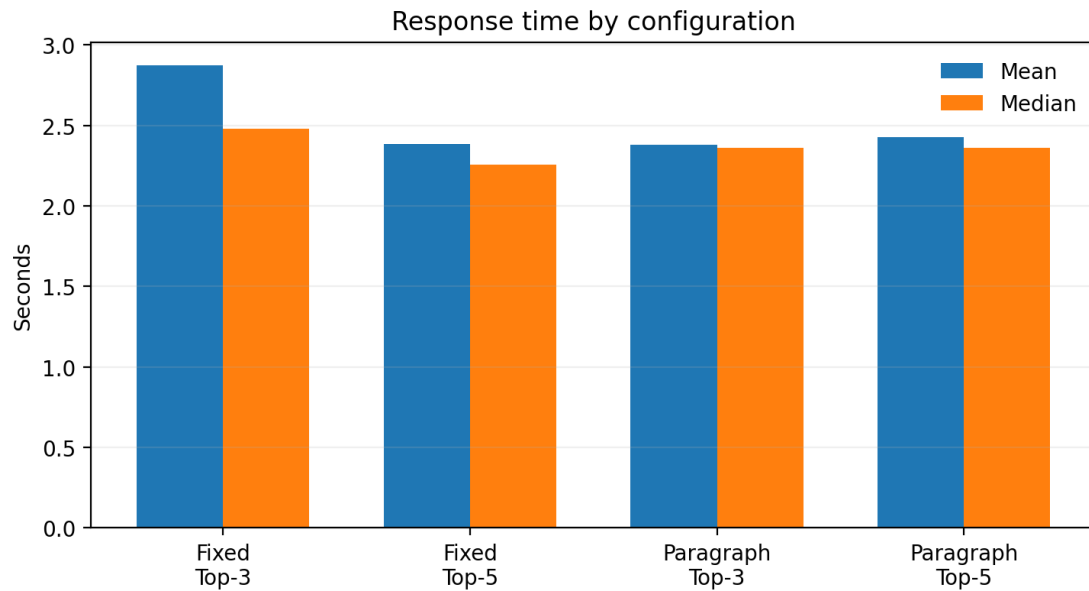


Figure 6. Mean and median response times for all 30 questions in each configuration.

The timing results do not establish a meaningful speed advantage for top-5. The fixed_top_3 mean is affected by a single 14.745-second observation; its median is 2.481 seconds and its mean without that observation is approximately 2.465 seconds. Because each question/configuration pair was timed only once and external service latency varies, response-time differences should be treated as descriptive rather than conclusive.

5.3 Analysis by Research Question

RQ1 - Chunking Strategy

Both chunking strategies achieved 100% retrieval hit, so neither failed to locate an expected document in this test. Paragraph-aware chunking provided stronger first-rank placement: 22 of 24 expected sources were ranked first compared with 19 of 24 for fixed-size chunking. Fixed-size chunking, however, produced lower non-gold context and a higher literal keyword score. Because the keyword metric undercounted semantically acceptable answers, the evidence indicates a trade-off rather than a definitive winner.

RQ1 conclusion

Chunking strategy remains unresolved. Retain fixed_top_3 as a control and paragraph_top_3 as the leading alternative, then compare them with semantic and independent human grading.

RQ2 - Retrieval Depth

Top-3 and top-5 both retrieved the expected source for every answerable question. Top-5 therefore added no retrieval recall in this run. It did add approximately 1.67 chunks per answerable question and increased non-gold-source context by 15.3 percentage points. Top-3 is therefore the strongest current baseline because it preserves retrieval recall while reducing context volume and potential distraction.

RQ3 - Abstention

All four configurations correctly handled the six unsupported questions after abstention-text normalization. This is encouraging evidence that the 0.35 relevance threshold and grounded prompt can reduce unsupported answers in the selected test set. The unanswerable set is small, however, and should be expanded with more difficult near-miss questions and threshold sweeps before stronger claims are made.

6. Discussion, Benefits, and Challenges

6.1 What the Prototype Demonstrates

The prototype demonstrates that a document Q&A workflow can be made substantially more inspectable without requiring a custom language model. It records which source was retrieved, where it ranked, what score it received, whether it passed the threshold, which passages were sent to generation, and what source labels appeared in the answer. This auditability is the main project contribution because it supports diagnosis rather than treating the final response as a black-box outcome.

The results also demonstrate that retrieval coverage and context quality are different objectives. Once the correct source is already present, increasing retrieval depth can increase the quantity of context without increasing useful evidence. This observation is consistent with research showing that language models do not always exploit long contexts robustly [3].

6.2 Error Analysis and Validity of the Answer Metric

There were 21 answerable configuration cases in which retrieval hit and citation-label inclusion succeeded but the automated keyword score failed. An author audit reviewed each question, expected phrase list, and generated response. All 21 responses conveyed the requested meaning. The audit was not independent or blinded, so it should not be interpreted as a final correctness study; however, it clearly shows that the strict metric substantially undercounts acceptable answers.

Failure type	Expected phrase(s)	Generated wording	Interpretation
Abbreviation expansion	small configurable RAG system	small configurable Retrieval-Augmented Generation system	Equivalent meaning; literal substring fails.
Synonym / paraphrase	safe; healthful; working conditions	a safe and healthful workplace	Equivalent wording.
Grammatical form	assigns tasks	assignment of tasks	Same fact expressed in noun form.
Phrase order / formulation	file a complaint; on-site inspection	file a written complaint and ask OSHA to inspect the workplace	Requested action is clearly answered.
Gold-answer scope mismatch	October 15; December 7; January 1	October 15 to December 7	Question asks when enrolment occurs; January 1 is a coverage start date.

Table 11. Examples of wording-sensitive grading failures identified during manual audit.

The practical implication is to rename the automated result as keyword coverage rather than answer correctness. Future evaluation should separate factual correctness, completeness, faithfulness to evidence, citation entailment, and conciseness. Expected-answer fields should also include only facts required by the question, and semantic grading should be calibrated against a small independently reviewed sample.

6.3 Benefits and Learning Outcomes

The work term produced both a functioning prototype and a stronger evaluation mindset. Technically, the project connected document extraction, metadata preservation, chunking, vector retrieval, relevance thresholds, grounded prompting, citations, abstention, and experiment logging in one system. From a software-engineering perspective, the work reinforced modular design, separation of concerns, reproducible CSV outputs, and evidence-based debugging.

The most important professional lesson was that evaluation design is part of system design. A system can retrieve the correct evidence and produce a reasonable answer while a poorly aligned metric still reports failure. The project therefore strengthened the ability to question measurements, communicate limitations, and distinguish an implementation problem from an evaluation artifact. It also connected academic concepts from information retrieval, NLP, software engineering, testing, and responsible AI to a practical application.

6.4 Problems, Challenges, and Mitigations

Challenge	Mitigation / lesson
PDF extraction quality	Focus on text-based PDFs, inspect extracted text, and preserve page/source metadata so retrieval errors can be traced.
Scope expansion	Freeze the minimum viable scope: one local user, three text-based file types, no production deployment or model training.
Context noise	Compare top-3 and top-5 directly and measure chunks sent and non-gold context rather than assuming more context is better.
Threshold calibration	Use one fixed threshold for controlled comparison; identify threshold sweeps as future work.
Literal answer scoring	Perform a manual error audit and reinterpret the metric as keyword coverage rather than semantic correctness.
API latency variability	Report timing descriptively, highlight the outlier, and recommend repeated measurements for future work.
Reproducibility metadata	Document that model identifiers were missing from exported evaluation artifacts and recommend capturing configuration metadata automatically in each run.

Table 12. Project challenges, mitigations, and lessons learned.

6.5 Limitations

- The evaluation contains 30 questions and six unanswerable cases. This is suitable for a prototype comparison but too small for broad generalization.
- The automated answer metric is wording-sensitive and does not directly measure semantic correctness or claim-level faithfulness.
- Citation-label inclusion verifies the presence of a displayed source label but not whether every generated claim is entailed by the cited passage.
- Non-gold-source context uses one designated expected source and may label useful supporting passages as non-gold.
- The corpus is limited to text-based files; advanced OCR, scanned documents, complex tables, and diagrams are outside scope.
- Only one embedding/vector-search setup and one generation configuration were evaluated, and exact model identifiers were not preserved in the exported evaluation artifacts.
- Latency measurements are single observations affected by external-service variability.
- The manual semantic audit was performed by the author and is therefore not independent or blinded.

7. Recommendations and Future Work

7.1 Recommended Current Baseline

Use $\text{top}_k = 3$ as the current retrieval baseline for this corpus. Top-3 preserved complete retrieval hit while sending fewer passages and less non-gold-source context than top-5. Continue comparing fixed_top_3 and paragraph_top_3 rather than declaring a final chunking winner until semantic and human evaluation can determine whether stronger first-rank placement or lower non-gold context matters more for final answer quality.

7.2 Improve Evaluation Quality

Replace the strict phrase checker as the primary correctness measure with a multidimensional rubric covering factual correctness, completeness, faithfulness to retrieved evidence, citation support, and conciseness. A small independently graded sample should be used to calibrate any LLM-assisted evaluator. Threshold values should also be evaluated on a development set so that precision/recall and abstention trade-offs are visible.

7.3 Adaptive Retrieval and Reranking

A practical extension is to begin with three retrieved passages and request more only when the initial evidence is weak, incomplete, or distributed across multiple document sections. Reranking can then prioritize the strongest passages before generation. This design would preserve the context-efficiency benefit observed in the current experiment while still allowing deeper retrieval when needed.

7.4 Expand Document Support and Production Readiness

Future document support should include OCR for scanned PDFs, stronger table and diagram handling, multilingual evaluation, and a broader mix of document styles. Production deployment should be treated as a separate engineering stage requiring authentication, privacy controls, encryption, access control, monitoring, cost limits, audit logging, and security testing.

7.5 Improve Reproducibility

Each evaluation run should automatically record exact embedding and generation model identifiers, service/provider versions, temperature and decoding settings, index configuration, library versions, and relevant environment metadata. This information should be stored beside the result CSV files so that later comparisons can distinguish algorithmic changes from environment changes.

8. Conclusion

Ask My Documents successfully implemented and evaluated a transparent RAG prototype for uploaded documents. In the selected test set, every configuration retrieved an expected source for all 24 answerable questions, included displayed source labels, and correctly abstained on the six unsupported questions. These results indicate that the core evidence-first workflow is functioning as intended within the scope of the prototype.

The strongest reliable finding concerns retrieval depth. Top-3 preserved complete retrieval hit while sending less context and less non-gold-source material than top-5. The evidence for chunking strategy is mixed: paragraph-aware chunking ranked the expected source first more often, while fixed-size chunking sent less non-gold context and scored higher under a literal phrase checker. Because that checker misclassified semantically acceptable responses, the report does not declare a definitive chunking winner.

The project demonstrates that trustworthy document question answering depends on more than fluent generation. It requires visible evidence, inspectable retrieval decisions, appropriate abstention, and evaluation methods that measure the intended behaviour. The next stage should improve semantic grading, expand the question set and document types, and capture complete model and environment metadata.

Final message

Trustworthy document AI needs evidence and transparent evaluation, not only fluent text.

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Project and Evaluation Artifacts

- Ask My Documents source modules: `app.py`; `rag/extractors.py`; `rag/chunking.py`; `rag/embeddings.py`; `rag/retriever.py`; `rag/generator.py`; `rag/prompts.py`; `rag/evaluator.py`.
- Evaluation outputs: `evaluation_results.csv`; `evaluation_chunk_details.csv`; `evaluation_digest_simple.csv`.
- `Project_Proposal.docx`, Ask My Documents project proposal, Summer 2, 2026.

Appendices

Appendix A - Reproducibility

The evaluation can be rerun from the project root using the following commands. The final submission should also record exact embedding and generation model identifiers, provider/service versions, and relevant environment variables. These identifiers were not preserved in the supplied evaluation exports and should be copied from the active project configuration if available.

```
PYTHONPATH=. .venv/bin/python -m rag.evaluator
.venv/bin/python -m pytest -q
```

Appendix B - Output Files

File	Rows	Columns	Purpose
evaluation_results.csv	120	26	One row per question and configuration; primary metrics and answer outputs.
evaluation_chunk_details.csv	480	20	One row per retrieved chunk; rank, score, threshold, source, and context analysis.
evaluation_digest_simple.csv	125	19	Human-readable configuration and per-question summary.

Table 13. Evaluation output files.

Appendix C - Corpus Inventory

Document	Type	Sections	Words	Fixed chunks	Paragraph chunks
CFPB FCRA Summary of Rights.pdf	PDF	4	1,496	15	14
CFPB Understand Your Credit Report.pdf	PDF	3	1,118	11	11
CFPB Understand Your Credit Score.pdf	PDF	2	776	8	7
FEMA Emergency Operations Plans CPG101.pdf	PDF	139	42,864	429	423
Medicare and You 2026.pdf	PDF	127	47,974	480	464
NIST AI Risk Management Framework.pdf	PDF	48	15,911	159	153
OSHA Employer Rights Responsibilities Inspection.pdf	PDF	27	4,836	49	52
OSHA Workers Rights.pdf	PDF	28	7,051	71	73
Project Proposal.docx	DOCX	39	1,056	11	11

Table 14. Evaluation corpus inventory.

Appendix D - Evaluation Corpus Sources

- Consumer Financial Protection Bureau - A Summary of Your Rights Under the Fair Credit Reporting Act.
- Consumer Financial Protection Bureau - Understand Your Credit Report.
- Consumer Financial Protection Bureau - Understand Your Credit Score.
- Federal Emergency Management Agency - Developing and Maintaining Emergency Operations Plans: Comprehensive Preparedness Guide 101.
- Centers for Medicare & Medicaid Services - Medicare & You 2026.
- National Institute of Standards and Technology - Artificial Intelligence Risk Management Framework (AI RMF 1.0).
- Occupational Safety and Health Administration - Employer Rights and Responsibilities Following an OSHA Inspection.
- Occupational Safety and Health Administration - Workers Rights.
- Ask My Documents - Project Proposal.